

Exercise Sheet 11 - Solutions

Predicate Logic

1. Here is a constructive Natural Deduction proof of $(S_1) \rightarrow (S_2) \rightarrow \forall x. \neg 0 < x$:

$$\begin{array}{c}
 \overline{S_2}^2 \\
 \hline
 \frac{\forall y. 0 < y \rightarrow 0 \leq \text{succ}(y)}{0 < x \rightarrow 0 \leq \text{succ}(x)} \quad [\forall E] \quad \overline{0 < x}^3 \\
 \hline
 0 \leq \text{succ}(x) \quad [\rightarrow E] \quad \overline{S_1}^1 \\
 \hline
 \neg 0 \leq \text{succ}(x) \quad [\forall E] \\
 \hline
 \frac{\perp}{\neg 0 < x}^3 \quad [\neg I] \\
 \hline
 \frac{\neg 0 < x}{\forall x. \neg 0 < x} \quad [\forall I] \\
 \hline
 \frac{\forall x. \neg 0 < x}{(S_2) \rightarrow \forall x. \neg 0 < x}^2 \quad [\rightarrow I] \\
 \hline
 \frac{(S_2) \rightarrow \forall x. \neg 0 < x}{(S_1) \rightarrow (S_2) \rightarrow \forall x. \neg 0 < x}^1 \quad [\rightarrow I]
 \end{array}$$

2. Here is a constructive Natural Deduction proof of $(S_3) \rightarrow (S_4) \rightarrow \forall x. \neg \text{succ}(x) \leq 0$:

$$\begin{array}{c}
 \overline{S_4}^2 \\
 \hline
 \frac{\forall y. \text{succ}(x) \leq y \rightarrow x < y}{\text{succ}(x) \leq 0 \rightarrow x < 0} \quad [\forall E] \quad \overline{\text{succ}(x) \leq 0}^3 \\
 \hline
 \frac{x < 0}{\exists x. x < 0} \quad [\exists I] \quad \frac{\text{succ}(x) \leq 0}{\neg \exists x. x < 0}^1 \\
 \hline
 \perp \quad [\neg E] \\
 \hline
 \frac{\perp}{\neg \text{succ}(x) \leq 0}^3 \quad [\neg I] \\
 \hline
 \frac{\neg \text{succ}(x) \leq 0}{\forall x. \neg \text{succ}(x) \leq 0} \quad [\forall I] \\
 \hline
 \frac{\forall x. \neg \text{succ}(x) \leq 0}{(S_4) \rightarrow \forall x. \neg \text{succ}(x) \leq 0}^2 \quad [\rightarrow I] \\
 \hline
 \frac{(S_4) \rightarrow \forall x. \neg \text{succ}(x) \leq 0}{(S_3) \rightarrow (S_4) \rightarrow \forall x. \neg \text{succ}(x) \leq 0}^1 \quad [\rightarrow I]
 \end{array}$$

3. Here is a constructive Natural Deduction proof of $(S_3) \rightarrow (S_4) \rightarrow \neg \exists x. \text{succ}(x) \leq 0$:

$$\begin{array}{c}
 \overline{S_4}^2 \\
 \hline
 \frac{\forall y. \text{succ}(x) \leq y \rightarrow x < y}{\text{succ}(x) \leq 0 \rightarrow x < 0} \quad [\forall E] \quad \overline{\text{succ}(x) \leq 0}^3 \\
 \hline
 \frac{x < 0}{\exists x. x < 0} \quad [\exists I] \quad \frac{\text{succ}(x) \leq 0}{\neg \exists x. x < 0}^1 \\
 \hline
 \frac{\exists x. x < 0}{\neg \exists x. \text{succ}(x) \leq 0}^3 \quad \frac{\neg \exists x. x < 0}{\neg \exists x. \text{succ}(x) \leq 0}^1 \\
 \hline
 \perp \quad [\neg E] \\
 \hline
 \frac{\perp}{\neg \exists x. \text{succ}(x) \leq 0}^3 \quad [\neg I] \\
 \hline
 \frac{\neg \exists x. \text{succ}(x) \leq 0}{(S_4) \rightarrow \neg \exists x. \text{succ}(x) \leq 0}^2 \quad [\rightarrow E] \\
 \hline
 \frac{(S_4) \rightarrow \neg \exists x. \text{succ}(x) \leq 0}{(S_3) \rightarrow (S_4) \rightarrow \neg \exists x. \text{succ}(x) \leq 0}^1 \quad [\rightarrow I]
 \end{array}$$

4. For example, the models

- $M_1 = \langle \mathbb{N}, \langle 0, \langle n \rangle \mapsto n + 1 \rangle, \langle \{ \langle n, m \rangle \mid n < m \}, \{ \langle n, m \rangle \mid n = m \} \rangle \rangle$ (because for example $0 < 1$ but they are not equal)
- and $M'_1 = \langle \{0\}, \langle 0, \langle n \rangle \mapsto n \rangle, \langle \{ \langle n, m \rangle \mid \text{True} \}, \emptyset \rangle \rangle$

are models of $\exists x. \exists y. x < y \wedge \neg x \leq y$;

5. For example, the models

- $M_2 = \langle \mathbb{N}, \langle 0, \langle n \rangle \mapsto n + 1 \rangle, \langle \{ \langle n, m \rangle \mid n < m \}, \{ \langle n, m \rangle \mid n \leq m \} \rangle \rangle$ (because any two numbers that are $<$ are also $\leq$)
- and $M'_2 = \langle \{0\}, \langle 0, \langle n \rangle \mapsto n \rangle, \langle \emptyset, \{ \langle n, m \rangle \mid \text{True} \} \rangle \rangle$

are models of $\neg \exists x. \exists y. x < y \wedge \neg x \leq y$.